

SimMat 2026

Welcome Session



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Guacolda-Leftraru Epu Supercomputer

The National Laboratory for High Performance Computing (NLHPC) is the Chilean national supercomputing center. It is hosted by the Center for Mathematical Modeling (CMM) at the Faculty of Physical and Mathematical Sciences of the University of Chile and provides computing resources to researchers throughout the country.



Accessing the Leftraru Cluster

Leftraru runs on the Linux operating system. Users can connect remotely through the SSH protocol.

Open a terminal and run the following command, replacing **USERNAME** with your NLHPC username:

```
ssh -p 4603 -l USERNAME leftraru.nlhpc.cl
```

The first time you connect, type **yes** to confirm the host and then enter your password.

Linux and Mac users

Linux and macOS users can save the connection details in the `~/.ssh/config` file. This allows you to connect to Leftraru using a shorter command.

Open or create the configuration file:

```
vim ~/.ssh/config
```

Add the following entry, replacing **USERNAME** with your NLHPC username:

```
Host leftraru
    HostName leftraru.nlhpc.cl
    User USERNAME
    Port 4603
    UpdateHostKeys no
```

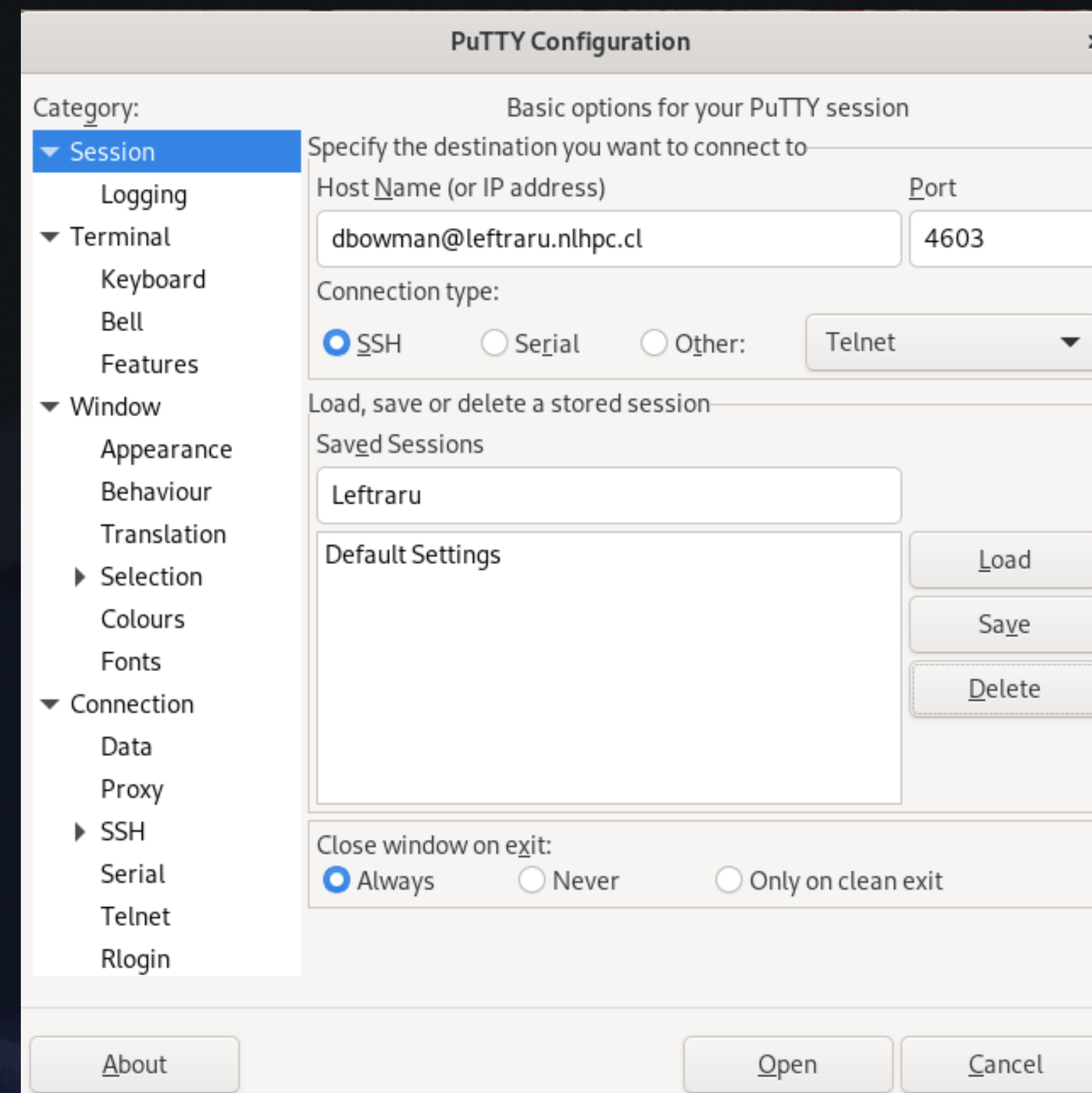
After saving the file, connect by running:

```
ssh leftraru
```


Window users

Windows users can connect to Leftraru using PuTTY as an SSH client.

In the PuTTY configuration window, enter:



Basic Unix Commands

Leftrararu uses a Linux environment. The following commands will help you navigate and manage files from the terminal.

- `ls` List files and directories
- `ls -ltrh` List files with details and human-readable sizes, ordered from oldest to newest
- `pwd` Show the current directory (path)
- `cd directory` Move/enter to another directory
- `cp file` Copy a file or directory
- `mkdir` Create a directory
- `mv` Move a file or directory
- `rm` Delete a file or folder
- `cat` Display the contents of a file
- `less` Read a file page by page

Downloading and Updating the Course Materials

The course materials are available in the SimMat 2026 GitHub repository.

First time: clone the repository

```
git clone https://github.com/simmat-uchile/MiniSchool_SimMat_2026.git
```

Later: update your local copy

From inside the repository, run:

```
git pull
```


Hands-on 0

1. Connect to Leftraru

Open a terminal on your local computer and connect using:

```
ssh -p 4603 -l USERNAME leftraru.nlhpc.cl
```

Replace USERNAME with your NLHPC username.

When using an SSH alias, simply run:

```
ssh leftraru
```


Hands-on 0

2. Access the course repository

The first time you access the course materials, clone the repository:

```
git clone https://github.com/simmat-uchile/MiniSchool_SimMat_2026.git
```

Enter the Hands-on 0 directory:

```
cd /home/courses/studentXX/MiniSchool_SimMat_2026/Exercises/Day_1/Welcome
```


Hands-on 0

3. Inspect the files

Check your current location:

```
pwd
```

List the files:

```
ls -ltrh
```

The directory should contain:

1. particle_in_box.py
2. run.sh

Hands-on 0

4. The physical problem

For a particle confined in a one-dimensional box of length (L), the normalized wave functions are:

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad 0 \leq x \leq L.$$

The Python script plots the first three quantum states:

$$n = 1, \quad n = 2, \quad n = 3.$$

Hands-on 0

5. Submit the job

Submit the calculation using:

```
sbatch run.sh
```

Check the status of your jobs:

```
squeue
```

Typical job states include:

PD Pending

CG Completing

R Running

Hands-on 0

6. Transfer the figure to your computer

Open a new terminal on your local computer (Do not run the following command from inside Leftrarú):

```
rsync -avz -e "ssh -p 4603 -o PreferredAuthentications=password -o PubkeyAuthentication=no" studentXX@leftrarú.nlhpc.cl:/home/courses/studentXX/MiniSchool_SimMat_2026/Exercises/Day_1/Welcome .
```

```
rsync -avzP leftrarú:~/MiniSchool_SimMat_2026/Exercises/Day_1/Welcome/particle_in_box.png .
```


Hands-on 0

6. Open the figure on your computer

